

Article

Multi-Objective Collaborative Optimization of Distribution Networks with Energy Storage and Electric Vehicles Using an Improved NSGA-II Algorithm

Runquan He ^{1,*}, Jiayin Hao ¹, Heng Zhou ¹ and Fei Chen ²

¹ Maoming Power Supply Bureau, Guangdong Power Grid Co., Ltd., Maoming 525000, China; haojiayin@mm.gd.csg.cn (J.H.); zhouheng@mm.gd.csg.cn (H.Z.)

² Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China; chen_fei1987@sjtu.edu.cn

* Correspondence: herunquan0303@163.com

Abstract

Grid-based distribution networks represent an advanced form of smart grids that enable modular, region-specific optimization of power resource allocation. This paper presents a novel planning framework aimed at the coordinated deployment of distributed generation, electrical loads, and energy storage systems, including both dispatchable and non-dispatchable electric vehicles. A three-dimensional objective system is constructed, incorporating investment cost, reliability metrics, and network loss indicators, forming a comprehensive multi-objective optimization model. To solve this complex planning problem, an improved version of the NSGA-II is employed, integrating hybrid encoding, feasibility constraints, and fuzzy decision-making for enhanced solution quality. The proposed method is applied to the IEEE 33-bus distribution system to validate its practicality. Simulation results demonstrate that the framework effectively addresses key challenges in modern distribution networks, including renewable intermittency, dynamic load variation, resource coordination, and computational tractability. It significantly enhances system operational efficiency and electric vehicles charging flexibility under varying conditions. In the IEEE 33-bus test, the coordinated optimization (Scheme 4) reduced the expected load loss from 100×10^{-4} yuan to 51×10^{-4} yuan. Network losses also dropped from 2.7×10^{-4} yuan to 2.5×10^{-4} yuan. The findings highlight the model's capability to balance economic investment and reliability, offering a robust solution for future intelligent distribution network planning and integrated energy resource management.

Keywords: energy storage; distribution network; optimal scheduling; coordinated response; NSGA-II; grid-based distribution network



Academic Editors: Jiachen Gao, Tianyan Jiang, Yaqi Fang and Jinxin Cao

Received: 2 September 2025

Revised: 28 September 2025

Accepted: 28 September 2025

Published: 2 October 2025

Citation: He, R.; Hao, J.; Zhou, H.; Chen, F. Multi-Objective Collaborative Optimization of Distribution Networks with Energy Storage and Electric Vehicles Using an Improved NSGA-II Algorithm. *Energies* **2025**, *18*, 5232. <https://doi.org/10.3390/en18195232>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

With the increasing popularity of energy storage (ES), electric vehicles (EVs), and renewable energy generation equipment such as wind turbines and photovoltaic systems, the current distribution network is rapidly transitioning into a multi-attribute composite system [1,2]. As the penetration of distributed energy sources increases, modern active distribution networks are adopting a networked layout, forming several autonomous unit groups through topology reconfiguration [3]. This decoupled architecture effectively achieves the hierarchical utilization and dynamic allocation of regional energy, significantly enhancing system resilience while improving asset utilization [4]. Each grid group has

relatively independent operational characteristics, but the groups are interconnected and influence each other, requiring an efficient collaborative optimization scheduling mechanism to coordinate their operation [5,6]. The integration of devices introduces a series of impacts on the distribution network, including changes in system power flow and effects on grid frequency and voltage [7]. Therefore, analyzing the characteristics of these devices and establishing mathematical models to study the operation of the distribution network has important economic and safety benefits.

In recent years, the multi-objective scheduling and configuration optimization of distribution networks have been widely studied. For example, reference [7] proposed a hierarchical electric vehicle charging scheduling strategy based on NSGA-II, which can reduce user electricity costs while improving the balance of system operation. Reference [8] used the improved particle swarm optimization method to carry out multi-objective collaborative optimization of active distribution networks, showing strong advantages in voltage stability and load peak reduction. In addition, reference [9] introduced distributed robust optimization into the multi-objective planning of distribution networks, significantly reducing the uncertainty risk of the scheme and improving the economic efficiency. Reference [10] studied the energy storage capacity optimization problem in the microgrid scenario, and comprehensively considered the coordinated scheduling of combined heat and power and electric vehicles, proving the important role of multi-energy coupling in improving operational flexibility. Reference [11] analyzed the impact of electric vehicle access mode on the static voltage stability margin of the distribution network and found that different access methods will significantly affect the load absorption capacity of the system. Reference [12] systematically reviewed the multi-objective optimization research of multi-energy systems and electric vehicle charging stations, pointing out that existing research still has deficiencies in user behavior modeling and grid-transportation network coupling. Reference [13] used the NSGA-II method to model the energy storage site selection and capacity configuration in active distribution networks, and verified the applicability of intelligent algorithms in solving non-convex multi-objective problems. Reference [14] proposed a safe energy strategy optimization method for multi-energy microgrid control, which effectively addressed the uncertainty problem caused by wind and solar output fluctuations. Reference [15] established a multi-objective optimization model for the short-term scheduling problem of a cascaded hydropower-wind power-photovoltaic-thermal power and pumped storage system, providing a reference for the coordinated operation of cross-energy systems. Reference [16] systematically analyzed the computational complexity of NSGA-II and other multi-objective evolutionary algorithms, laying the foundation for the selection of algorithms in subsequent power system optimization problems. These studies show that multi-objective intelligent optimization methods have good performance in terms of convergence and diversity, but most of the results are still at the simulation level and lack verification in actual engineering scenarios.

In summary, existing research has achieved considerable results in the optimization of energy storage, electric vehicles, and distributed generation (DG). This paper aims to improve the operational efficiency and performance of the distribution network by fully utilizing the adjustment capabilities of ES, EV, and DG to optimize the operation of the distribution network [17,18]. Simulation tests based on the standard IEEE 33-node system show that the proposed modeling method has significant advantages in solution accuracy. Engineering verification from a regional grid renovation project further confirms its superiority in computational efficiency. The results demonstrate strong engineering application value. This study highlights the incremental contribution of the improved NSGA-II by demonstrating faster convergence and better Pareto diversity than standard NSGA-II, PSO, and distributed robust optimization methods.

2. Basic Equipment Models

2.1. Distributed Generation Output Models

The wind-turbine output model is given in Equation (1), while the photovoltaic output model is shown in Equation (2) [14,19].

$$P_{WT} = \begin{cases} 0 & 0 \leq v < v_{ci}, v_{co} < v \\ P_r - WT \left(\frac{v^3 - v_{ci}^3}{v_r^3 - v_{ci}^3} \right) & v_{ci} \leq v < v_r \\ P_{r-PV} - WT & v_r \leq v \leq v_{co} \end{cases} \quad (1)$$

$$P_{PV} = n_{PV} P_{r-PV} (R_c / R_r) [1 + k(T_c - T_r)] \quad (2)$$

where P_{WT} is the output power of the wind turbine; P_{r-WT} is the wind turbine's rated power; v_{ci} , v_{co} , and v_r denote the cut-in, cut-out, and rated wind speeds, respectively; P_{PV} is the photovoltaic array output power; n_{PV} is the number of photovoltaic cells; P_{r-PV} is the rated output power of the photovoltaic cells; R_c and R_r are the actual and rated irradiance; k is the power temperature coefficient; and T_c and T_r are the actual and standard temperatures, respectively.

2.2. Non-Dispatchable Electric Vehicles Load Model

Electric vehicles can be connected to the grid in two different ways. The first is the dispatchable mode, in which the electric vehicle can both charge and discharge, functioning like an energy storage device. The second is the non-dispatchable mode, in which the electric vehicle only draws power from the grid for charging [20]. For non-dispatchable electric vehicles, their daily energy consumption depends on the daily driving distance and follows the normal distribution given by Equation (3) [11]. By sampling Equation (3) for each electric vehicle, the daily driving distance can be obtained, and the daily energy consumption can be calculated as shown in Equation (4). Based on the driving distance, the remaining charge (SOC) of the electric vehicle can be determined using Equation (5). Finally, the resulting daily energy demand, that is, the required charging time per day, can be determined using Equation (6).

$$f(D_{EV}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(D_{EV} - \mu)^2}{2\sigma^2}\right) \quad (3)$$

$$E_{EV-consumption} = D_{EV} C_{EV} / D_{EVmax} \quad (4)$$

$$SOC_{EV} = (C_{EV} - E_{EV-consumption}) / C_{EV} \times 100\% \quad (5)$$

$$T_{EV-charge} = E_{EV-consumption} / P_{EV-charging} \quad (6)$$

where D_{EV} represents the daily driving distance of new energy vehicles; σ and μ are the standard deviation and mean of the normal distribution; $E_{EV-consumption}$ is the electric vehicle's daily energy consumption; D_{EVmax} is the maximum driving distance; C_{EV} is the battery capacity of the electric vehicle; SOC_{EV} is the remaining state of charge after the electric vehicle has driven is the state of charge remaining after the vehicle has travelled D_{EV} ; $T_{EV-charge}$ is the total charging time required by the electric vehicle per day; and $P_{EV-charging}$ is the charging power of the electric vehicle.

While the above formulation describes the daily energy demand and load profile for non-dispatchable EVs, both EV types in this study are modeled under a set of common default behavioral assumptions to ensure consistency and generality across different operating scenarios, as outlined below.

For non-dispatchable EVs, it is assumed that they operate solely as charging loads without any discharging capability. Their daily charging demand is derived from historical driving distance distributions, with arrival and departure times implicitly reflected in the aggregated load profile. For dispatchable EVs, it is assumed that they can both charge and discharge, subject to the following constraints:

- (1) SOC limits ($SOC_{min} \leq SOC(t) \leq SOC_{max}$);
- (2) Maximum charging/discharging power limits ($P_{c,max}, P_{d,max}$);
- (3) Availability windows aligned with typical residential/workplace charging behavior.

These assumptions are consistent with widely adopted EV modeling approaches in the literature while enabling the proposed framework to remain generalizable to different scenarios.

2.3. Energy Storage and Schedulable Electric Vehicles Charging–Discharging Model

The dispatchable EVs and ES have the same function and can be considered larger-capacity batteries; so, they can be described using the same model. The energy balance equations for charging/discharging are shown in Equations (7) and (8). The constraints on SOC and charging/discharging power are shown in Equations (9)–(11).

$$E_{es}(t) = E_{es}(t - \Delta t) + \eta P_{es-c} \Delta t \quad (7)$$

$$E_{es}(t) = E_{es}(t - \Delta t) - P_{es-f} \Delta t / \eta \quad (8)$$

$$SOC_{min} \leq SOC \leq SOC_{max} \quad (9)$$

$$P_{es-cmin} \leq P_{es-c} \leq P_{es-cmax} \quad (10)$$

$$P_{es-fmin} \leq P_{es-f} \leq P_{es-fmax} \quad (11)$$

where $E_{es}(t)$ is the energy stored in the ES at time t ; η is the charge/discharge loss coefficient of the ES; and P_{es-c} and P_{es-f} are the charging and discharging power of the ES, respectively. SOC_{max} and SOC_{min} define the upper and lower bounds of the state of charge. $P_{es-cmin}$ and $P_{es-cmax}$ set the lower and upper limits for charging power, while $P_{es-fmin}$ and $P_{es-fmax}$ are the lower and upper limits of discharging power.

3. Coordinated Response Model and Algorithm for Energy Storage and Electric Vehicles

3.1. Objective Function

The key to collaborative optimization lies in the reasonable allocation of DG, load, and energy storage devices both within and outside the distribution network. From the perspective of the power supply company, this paper models this complex issue as a multi-objective collaborative optimization problem, aiming to meet various constraints and seek efficient solutions. In the planning and construction of distributed power sources, the main differences lie in the power supply methods and investment costs. This makes the optimization model not only focus on economy but also consider the reliability of power supply. Through this comprehensive consideration, the optimization model in this manuscript strives to achieve the best balance between economy and reliability, providing scientific basis and guidance for the planning of grid-based distribution networks.

- (1) Investment cost F_m

The mathematical expression for the investment cost is:

$$\min F_m = \min \left[\sum_{p=1}^{N_{es}} C_{ES} \times P_{ES.p} + \sum_{i=1}^{N_{pv}} C_{pv} \times P_{pv.i} + \sum_{j=1}^{N_{wind}} C_{wind} \times P_{wind.j} + C_G \right] \quad (12)$$

where N_{es} denotes the number of energy-storage sites, N_{pv} the number of photovoltaic groups, N_{wind} the number of wind-turbine groups, and N_l the number of load groups. The capacity of the j -th wind group is $P_{wind \cdot j}$ with a per-unit investment cost C_{wind} ; the per-unit investment cost of storage is C_{ES} , and the capacity of the p -th storage group is $P_{ES \cdot p}$. Likewise, the capacity of the i -th PV group is $P_{pv \cdot i}$ with a unit-capacity investment cost C_{PV} . The distribution-network construction cost is expressed as $C_G = AS_G + b$, where S_G denotes the capacity scale of the distribution network, A represents the proportionality between the construction capacity and cost, and b represents the fixed cost of distribution network construction

(2) Expected Grid Energy Shortage E_m

The objective function is designed to quantify the reliability of user experience under different power supply modes. This paper not only quantifies power supply reliability but also provides a basis for optimization decisions. The method helps power companies identify potential risk points and take corresponding measures to improve the reliability of the distribution network system, ensuring that users always receive a stable power supply under different operating modes. In summary, focusing on the expected power shortage in the grid not only helps improve the overall performance of the power supply system but also provides users with a higher level of electricity security.

Assuming the failure rate of the energy storage devices in the distribution network is p_{es} , the internal line probability is $p_{Line-in}$, and the probability of the distribution network being in an islanding state is p_{island} , the probability of load loss within the distribution network can be expressed as follows:

$$p_{in} = p_{Line-in} + p_{es} \times p_{island} \quad (13)$$

The failure probability of external lines and the main grid fault probability are $p_{Line-out}$ and p_{system} , respectively. The probability of load loss in the external area is:

$$p_{on} = p_{Line-out} + p_{es} \times p_{system} \quad (14)$$

Therefore, the objective function for the expected load-loss value is:

$$\begin{aligned} \min E_m &= \min \{ p_{in} \times p_{L \cdot MG} + p_{out} \times (p_{TL} - p_{L \cdot MG}) \} \\ &= \min \left\{ p_{in} \times \sum_{x_{L \cdot i} \geq x_{ES}}^{x_{L \cdot N_1}} P_{L \cdot i} + p_{out} \times (P_{TL} - \sum_{x_{L \cdot i} \geq x_{ES}}^{x_{L \cdot N_1}} P_{L \cdot i}) \right\} \end{aligned} \quad (15)$$

where the total system load and the total load within the distribution network region are defined as p_{TL} and $p_{L \cdot MG}$, respectively; the node locations for the i -th group of load and energy storage are defined as $x_{L \cdot i}$ and x_{ES} ; the load value at the $x_{L \cdot i}$ node is defined as $p_{L \cdot i}$; $x_{ES} \sim N$ represents the distribution network region, where the number of nodes is defined as N ; and $x_{L \cdot i} \times x_{ES} \in [2, N]$ [10].

(3) Network Loss F_{loss}

In the optimization process, the impact of different access locations on network losses is first analyzed. By combining load characteristics and the generation characteristics of DG, the optimal access scheme is developed. By reasonably configuring the locations of DG, load, and energy storage devices, more efficient power transmission can be achieved, reducing the operational costs of the grid.

$$\min F_{loss} = \min \sum_{(i,j) \in [1 \cdot N]} G_{ij} (V_i^2 + V_j^2 - 2V_i V_j \cos \theta_{ij}) \quad (16)$$

where the admittance of branch ij is defined as G_{ij} ; V_i and V_j are the voltages at nodes i and j , respectively; and the voltage phase difference is defined as θ_{ij} .

(4) Operation and Maintenance Cost F_{om}

In addition to the investment cost, the operation and maintenance (O&M) cost is also considered to reflect the long-term economic impact of equipment deployment. The total planning cost is thus revised as follows:

$$F_m = F_{inv} + F_{om} \quad (17)$$

where F_{inv} is the investment cost defined in Equation (17), and F_{om} is the operation and maintenance cost, calculated as follows:

$$F_{om} = \sum_{j=1}^{N_{wind}} C_{wind}^{om} \cdot P_{wind,j} + \sum_{i=1}^{N_{pv}} C_{pv}^{om} \cdot P_{pv,i} + \sum_{p=1}^{N_{es}} C_{es}^{om} \cdot P_{es,p} \quad (18)$$

Here, C_{wind}^{om} , C_{pv}^{om} , C_{es}^{om} represent the annual O&M cost coefficients for wind turbines, photovoltaic panels, and energy storage systems, respectively.

3.2. Constraints

(1) Equality constraints:

① Power-flow equations:

$$\begin{aligned} P_i &= V_i \sum_{j=1}^n V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) \\ Q_i &= V_i \sum_{j=1}^n V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \end{aligned} \quad (19)$$

where P_i and Q_i are the active and reactive power injected at node i ; V_i and V_j are the voltage magnitudes at nodes i and j ; and G_{ij} and B_{ij} are the real and imaginary parts of the line admittance, respectively [10].

② Capacity constraints for DG and loads:

$$\begin{aligned} \sum_{i=1}^{N_{pv}} P_{pv,i} &= P_{TPV} \\ \sum_{j=1}^{N_{wind}} P_{wind,j} &= P_{TW} \end{aligned} \quad (20)$$

Load design capacity:

$$\sum_{d=1}^{N_L} P_{L,d} = P_{TL} \quad (21)$$

where P_{TPV} and P_{TW} represent the total design capacity of photovoltaic and wind power, respectively; $P_{L,d}$ is the load capacity of the d -th group, and P_{TL} is the total design capacity of the load.

(2) Inequality constraints

① Branch-power limits:

$$S_k \leq S_k^{\max} \quad (22)$$

where S_k is the apparent power on branch k ; $S_{\max}^k$ is the branch-capacity limit; $k \in [1, N_b]$ with N_b denoting the total number of branches.

② Output limits for distributed generators (DGs):

$$\begin{cases} P_{DG,i}^{\min} \leq P_{DG,i} \leq P_{DG,i}^{\max} \\ Q_{DG,i}^{\min} \leq Q_{DG,i} \leq Q_{DG,i}^{\max} \end{cases} \quad (23)$$

where the active and reactive power output of distributed generation i are defined as $P_{DG,i}$ and $Q_{DG,i}$, respectively; the upper and lower limits of the active power output and reactive power output of the DG are defined as $P_{DG,i}^{\max}$, $P_{DG,i}^{\min}$, $Q_{DG,i}^{\max}$ and $Q_{DG,i}^{\min}$.

③ Practical Constraints of Siting Distributed Energy Resources (DERs):

In addition to power-flow and capacity constraints, practical deployment constraints for DERs are considered to improve the engineering feasibility of the planning scheme. These include:

Spatial availability constraints: Only buses with sufficient installation area and low congestion are considered for DER deployment.

Short-circuit capacity limits of existing switchgear: The maximum allowable injection from DERs at each bus is limited to avoid exceeding short-circuit ratings.

Grid connection feasibility: DERs can only be installed at buses with pre-existing connection interfaces or low-cost upgrade potential.

To reflect these considerations, a binary feasibility indicator $\delta_i^{feas} \in \{0, 1\}$ is introduced for each candidate node i , where $\delta_i^{feas} = 1$ means the site is eligible for DER installation. The allocation variable at each bus is thus constrained as follows:

$$P_i^{DER} \leq \delta_i^{feas} \cdot P_i^{\max} \quad (24)$$

where $P_i^{\max}$ is the upper limit considering electrical and physical constraints.

4. Solution Procedure for the Configuration Scheme Based on the NSGA-II Algorithm

4.1. Planning Model and Control-Variable Encoding Strategy

Unlike the conventional NSGA-II, our improved version incorporates a hybrid-encoding scheme for siting and sizing variables, a feasibility-based constraint-handling approach to maintain solution validity, and a fuzzy-membership decision-making process for selecting the final compromise solution. The configuration problem of DG, load, and energy storage is transformed into an optimization model. The objective of the model is to achieve the minimum value of the objective function while satisfying both equality and inequality constraints. Through this transformation, the configuration of various resources can be systematically analyzed and optimized, thereby enabling the efficient operation of the power system. Specifically, the mathematical formulation of the optimization model is as follows:

$$\begin{cases} \min f(x_c, x_s) = \min\{f_1(x_c, x_s), f_2(x_c, x_s), f_3(x_c, x_s)\} \\ s.t. h_i(x_c, x_s) = 0, \quad i = 1, 2 \\ g_i(x_c, x_s) \leq 0, \quad i = 1, 2, \dots, 7 \end{cases} \quad (25)$$

where the state variables and independent control decision variables are defined as x_s and x_c , respectively; $f(x_c, x_s)$ is the optimization objective function; sub-objective functions such as investment cost, expected power shortage in the grid, and network loss are defined as $P_{DG,i}^{\max}$ and $P_{DG,i}^{\min}$, $Q_{DG,i}^{\max}$ and $Q_{DG,i}^{\min}$; the equality constraint is $h_i(x_c, x_s)$, and the inequality constraint is $g_i(x_c, x_s)$.

The mixed-encoding scheme for the control variables can be written as follows:

$$x = [T_{N_1}, L_{N_1}, P_{N_1}, T_{N_2}, L_{N_2}, P_{N_2}, \dots, T_{N_{ALL}}, L_{N_{ALL}}, P_{N_{ALL}}] \quad (26)$$

4.2. Multi-Objective Handling and Optimal-Solution Selection

In this study, an improved NSGA-II is developed to better handle the collaborative optimization problem involving both discrete siting and continuous sizing variables. The improvements include the following: (1) a hybrid-integer encoding scheme for mixed variable types; (2) a feasibility-priority constraint-handling mechanism; (3) a fuzzy-membership-based decision-making process for compromise solution selection; and (4) adaptive control of crossover and mutation rates. These modifications enhance convergence speed, feasible-solution ratio, and decision-making interpretability compared to the standard NSGA-II.

The flowchart shows the NSGA-II process. It starts by initializing the population and sorting individuals by non-domination. Offspring are generated through selection, crossover, and mutation. Parent and offspring populations are merged, sorted again, and evaluated using crowding distance. The best individuals are selected for the next generation. This repeats until the maximum generation is reached, yielding a Pareto-optimal solution set.

By using the NSGA-II algorithm, the optimized Pareto solution set can be effectively obtained [16,21]. To select the optimal solution from the high-quality solutions, this paper introduces the fuzzy membership function, which reflects the decision-maker's satisfaction with each optimization objective and is used to find the optimal solution [22]. In this process, based on Equation (24), the membership function value of the k -th non-dominated solution in the Pareto solution set in the i -th objective direction is quantified and solved. This method not only provides decision-makers with more intuitive optimization results but also offers a scientific basis for selecting the optimal solution. Through the evaluation of fuzzy membership, it better meets the needs of decision-makers, thereby achieving more reasonable and effective resource allocation [16].

Within this framework, the NSGA-II algorithm is highly regarded for its outstanding capability to handle multiple variables. The flowchart of the NSGA-II procedure is shown in Figure 1.

$$u_i^k = \begin{cases} 1, & F_i = F_i^{\min} \\ \frac{F_i^{\max} - F_i}{F_i^{\max} - F_i^{\min}}, & F_i^{\min} \leq F_i \leq F_i^{\max} \\ 0, & F_i = F_i^{\max} \end{cases} \quad (27)$$

where the lower and upper bounds of the objective function are defined as $F_i^{\min}$ and $F_i^{\max}$, respectively; the value of objective function i is defined as F_i . A hierarchical weighting strategy is established based on decision preference information, and a weighted fuzzy membership aggregation operation is applied to the Pareto front solution set, achieving the optimal solution selection in a multi-dimensional decision space. This paper uses an equal weighting method, which simplifies the complexity of weight setting and ensures that each objective function receives equal attention during the optimization process. The specific calculation is formulated as follows:

$$u^k = \frac{\sum_{i=1}^{N_{obj}} \lambda_i u_i^k}{\sum_{k=1}^{N_p} \sum_{i=1}^{N_{obj}} \lambda_i u_i^k}, \quad k = 1, 2, \dots, N_p \quad (28)$$

where N_p is the population size, λ_i is the weight assigned to the i -th objective function, N_{obj} is the total number of objectives, and u_i^k is the corresponding membership-degree value.

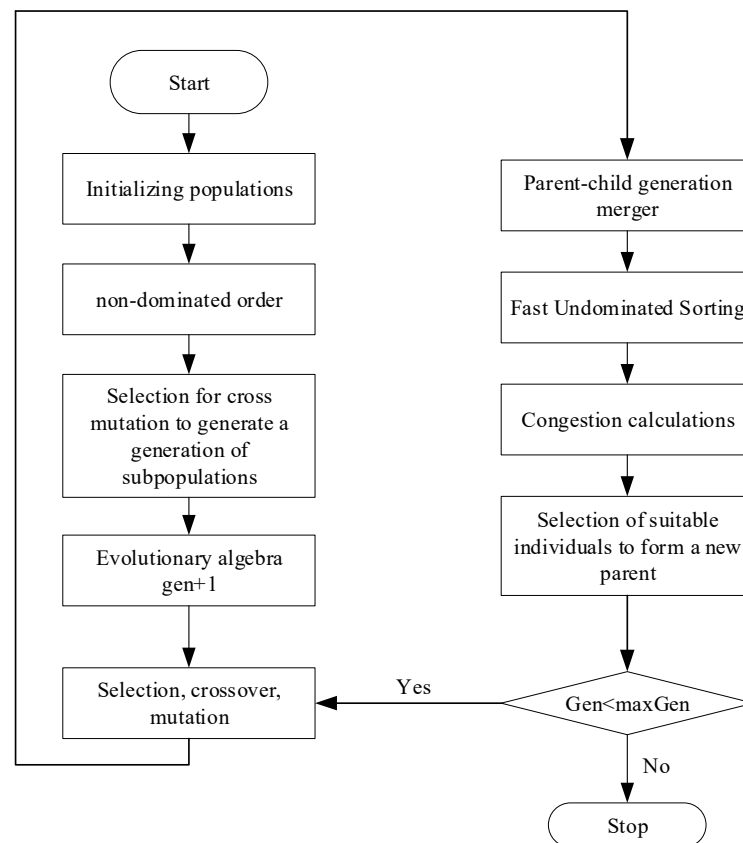


Figure 1. Flowchart of the NSGA-II Algorithm.

5. Case Study Analysis

5.1. Case Parameters

This paper validates the model through the IEEE 33-node distribution network test system, as shown in Figure 2, focusing on the longest feeder area, which covers 5 to 17 nodes, and conducts a grouping analysis of DG and the grid. The designed DG includes wind power with a capacity of 0.50 MW, photovoltaic with 1.10 MW, and load with 2.00 MW. Specific data can be found in Table 1.

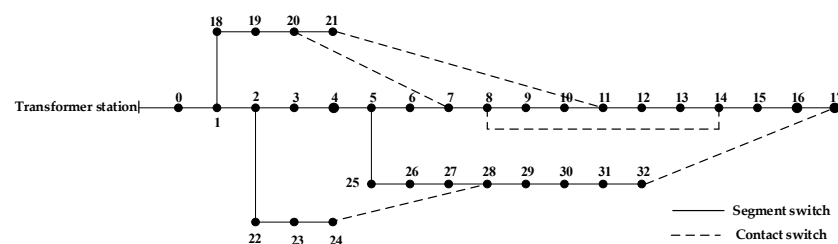


Figure 2. IEEE 33-Bus Distribution Network.

Table 1. Selected Parameter Values Used in the Calculations.

Parameter Name	Retrieve a Value	Parameter Name	Retrieve a Value
C_{ES}	¥130 million/MW	ε	0
C_{PV}	¥100 million/MW	λ	0.25
C_{wind}	¥100 million/MW	λ'	0.33
a	¥46.5 million/MW	$N_{l,max}$	3
b	¥5 million	$N_{pv,max}, N_{wind,max}$	3
p_{in}	0.0013	p_{out}	0.01

5.2. Comparison and Analysis of Computational Results

(1) Optimization Results and Discussion

The crossover and mutation factors are both set to 0.8, the population size and maximum number of evolutionary generations are set to 250 and 100, respectively, and the penalty factors $w_1, w_2, w_3 = 6.0 \times 10^{-3}$; $w_4, w_5 = 1.0 \times 10^{-3}$; $w_6, w_7 = 7.0 \times 10^{-3}$; and $w_8 = 1.0 \times 10^{-3}$. According to the optimization calculation, non-dominated sorting is performed, and the optimization results at the first level are selected as Pareto solutions. The surface formed by the Pareto solution set in space is shown in Figure 3. It can be seen from the figure that the Pareto solution set is relatively evenly and continuously distributed in the target space and can better cover the trade-off relationship between different objectives, indicating that the improved NSGA-II algorithm adopted has a strong ability to maintain solution set diversity and certain convergence performance.

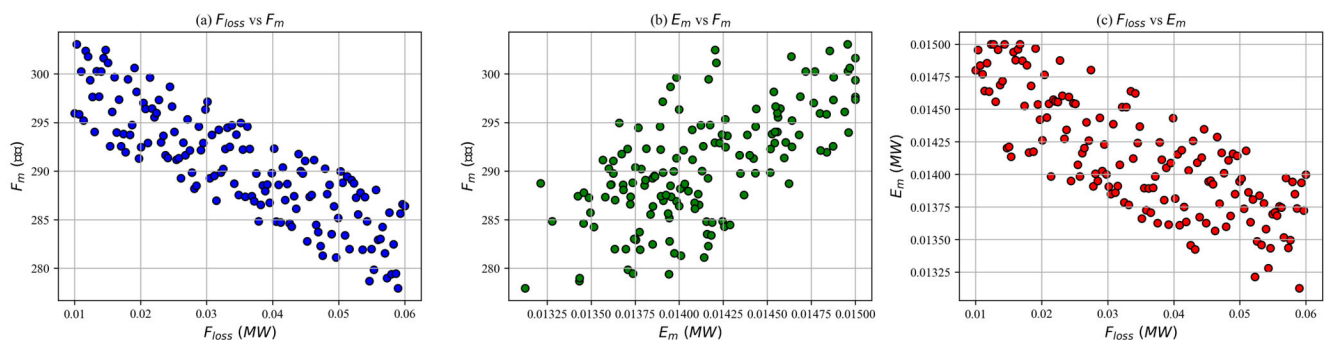


Figure 3. Pareto solution set.

The optimization results are selected as Pareto solutions, and the membership weight values are calculated using Equation (24) to evaluate the quality of each solution. A suitable compromise solution is then selected from these Pareto solutions to form the optimal configuration scheme. In this process, particular attention is given to the specific layout and capacity configuration of DG, load, and energy storage systems, with detailed information provided in Table 2. Through precise configuration, the system can ensure that load demands are met while optimizing resource utilization efficiency, thus enhancing the overall operational economy and reliability. Ultimately, this configuration scheme not only lays the foundation for the system's sustainable development but also provides strong support for future practical applications [15,23–26].

Table 2. Optimized Configuration Results.

Typology	Access Point Location	Quantitative/MW
Wind power	9	0.0159
Wind power	16	0.2299
Photovoltaic	6	0.4201
Photovoltaic	16	0.1496
Load	6	0.4121
Load	11	0.2201
Load	16	0.3610
Energy storage	10	0.1998

For comparison, a PSO algorithm was also applied to the same test system. Figure 4 shows that the improved NSGA-II converges much faster and stabilizes within about 40 generations, while PSO requires about 75 generations. As shown in Table 3, the improved NSGA-II also achieves a lower investment cost (167×10^{-4} yuan vs. 190×10^{-4} yuan),

lower expected load loss (51×10^{-4} yuan vs. 65×10^{-4} yuan), and reduced network loss (2.5×10^{-4} yuan vs. 3.1×10^{-4} yuan). Moreover, the IGD and spread values confirm that the proposed method produces more diverse and higher-quality Pareto solutions. These results highlight the advantage of the improved NSGA-II over PSO in both convergence speed and solution quality.

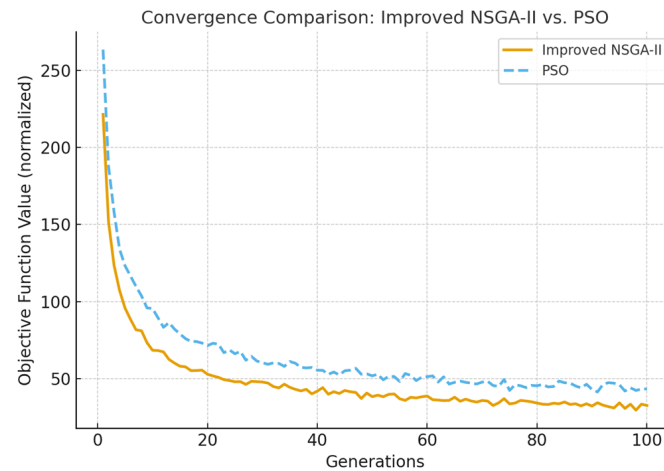


Figure 4. Comparison results with PSO.

Table 3. Performance comparison between improved NSGA-II and PSO.

Method	Convergence Generations	Fm (10^{-4} Yuan)	Em (10^{-4} Yuan)	Floss (10^{-4} Yuan)
Improved NSGA-II	40	167	51	2.5
PSO	75	190	65	3.1

In addition to the comparison with PSO, the improved NSGA-II was also compared with the standard NSGA-II under the same conditions. The results show that the improved NSGA-II achieves convergence within 40 generations, while the standard NSGA-II requires about 60 generations to reach a similar level of stability. Furthermore, the improved version provides a lower investment cost, lower expected loss, and reduced network loss compared with the standard NSGA-II. These improvements confirm that the proposed modifications enhance convergence speed and solution quality without increasing computational complexity.

(2) Comparison of Power-Supply Schemes

Scheme 1: DG is connected directly to the grid.

Scheme 2: Power is supplied solely by energy-storage units and schedulable EVs.

Scheme 3: The objective function E_m is clearly prioritized over the other two objectives, and the weighting vector is set to [0.3, 0.4, 0.3].

Scheme 4: A multi-objective coordinated-optimization mechanism is built, achieving globally optimal decisions via Pareto-front analysis.

According to the test data in Table 4, Scheme 1—using a DG-to-grid architecture and omitting any energy-storage system—reduces the initial investment by more than 50% compared with the other schemes.

Table 4. Comparison of Different Power-Supply Schemes.

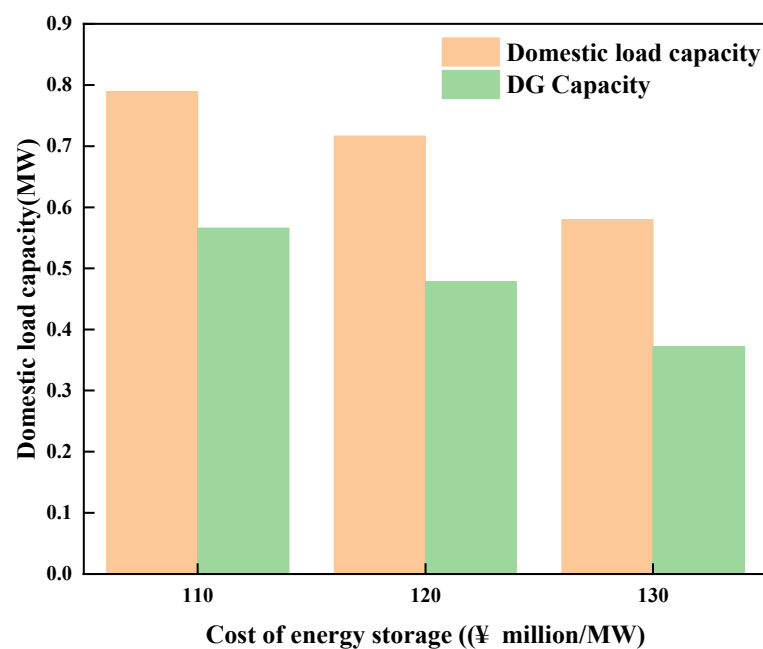
Program	Distribution Network Capacity /MW	F_m / (10^{-4} Million Yuan)	E_m / (10^{-4} Million Yuan)	F_{loss} / (10^{-4} Million Yuan)
1	—	80	100	2.7
2	1.8000	215	27	27
3	1.6613	196	33	12
4	1.1501	167	51	2.5

In Scheme 3, relative to Scheme 2, placing greater emphasis on supply reliability raises the ratio of the reliability index F_m to the expected load-loss value F_{loss} . This means that lowering the expected load-loss requires additional capital outlays and may simultaneously increase network losses.

The data from Table 5 shows that energy storage costs have a significant impact on the optimization results. The sensitivity test based on a 10% increase in energy storage costs indicates that the system configuration capacity and planning scale exhibit a synchronized contraction, forming a bidirectional coupling relationship. This phenomenon validates that the planning scheme is highly sensitive in terms of economic constraints. Therefore, a strengthened lifecycle cost management mechanism is required during the planning design phase. This will ensure the economic feasibility of the scheme (Figures 5 and 6).

Table 5. Comparison of Optimization Results under Different Energy-Storage Costs.

Energy Storage/(Million Yuan/MW)	Energy Storage Configuration		On-Net Load and DG		Off-Net Loads and DG	
	Placement	Quantitative/MW	Load/MW	DG/MW	Load/MW	DG/MW
110	9	0.2468	0.7901	0.5702	0.2097	0.2339
120	13	0.2311	0.7184	0.4811	0.2796	0.3236
130	10	0.1998	0.5812	0.3698	0.4197	0.4323

**Figure 5.** Comparison of In-Network Load and DG Capacity under Different Energy-Storage Costs.

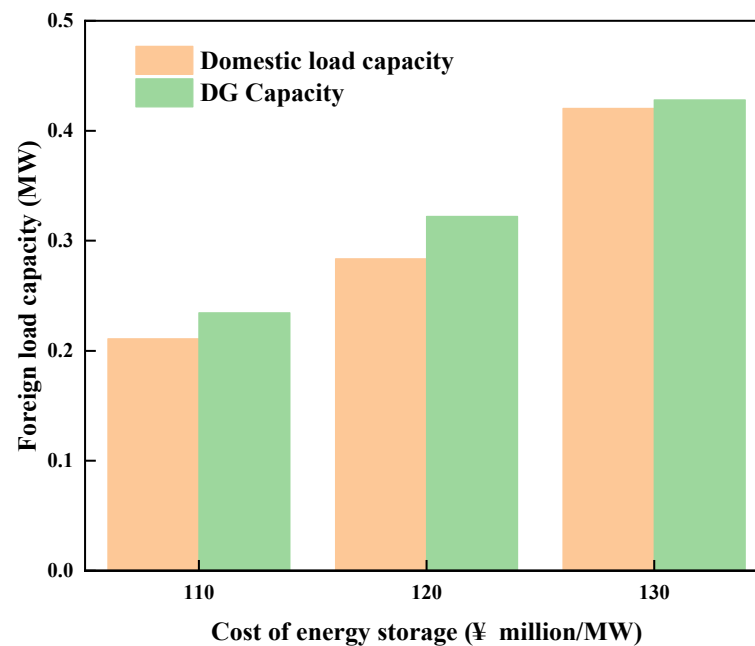


Figure 6. Comparison of Out-of-Network Load and DG Capacity under Different Energy-Storage Costs.

5.3. Sensitivity Analysis

To verify the robustness of the proposed framework under different operating points and access locations, this paper conducts two complementary sensitivity studies:

- (i) Post-evaluation sensitivity: The optimal configuration in Table 2 is fixed (the capacity and nodes of PV, WT, and ES remain unchanged), and only the operating point (daytime/nighttime, irradiation intensity, wind speed, and ES operation strategy) is changed, and the operability indicators are recalculated.
- (ii) Re-optimization sensitivity: When the ES access location (trunk/terminal) is changed, it is re-optimized as a planning variable, and the migration of the compromise solution and the change in indicators are compared.

Reference and setting instructions: PV output is linearly scaled with irradiation R_c according to Equation (2) and linearly corrected with temperature T_c ; wind power responds nonlinearly with wind speed v through the power curve according to Equation (1); ES charging and discharging are constrained by SoC and power limit according to Equations (7)–(11). Post-evaluation sensitivity maintains the investment cost F_{inv} constant, reporting only changes in expected power outage E_m and network loss F_{loss} (operating point changes do not change capital expenditures). Reoptimization sensitivity changes involve rerunning the improved NSGA-II after ES is connected to the busbar, allowing for capacity and location changes [26,27]. The same network and load as in Table 1 are used; “Baseline” corresponds to the operating point of the compromise solution in Table 2 (sunny day sunshine, rated near-rated wind speed, ES real-time strategy). Table 6 presents the results of the post-optimization sensitivity analysis.

S1/S2 reflects the primary effect of $P_{PV} \propto R_c$ and the secondary correction of the temperature coefficient in Equation (2); S3/S4 reflects the nonlinear power response caused by wind speed changes in Equation (1); S5/S6 corresponds to the impact of the two types of scheduling strategies on power flow distribution and voltage margin under Equations (7)–(11). Compared with the baseline, the cloudy and low wind speed scenarios cause E_m and F_{loss} to increase by approximately 5–20% and 6–28%, respectively, which is consistent with power flow physics; the real-time strategy reduces E_m by approxi-

mately 5–6% and F_{loss} by approximately 8% relative to the fixed strategy, which is in line with the expectation of “on-demand compensation”.

Table 6. Post-optimization sensitivity.

Scenario	Operating Setting	F_{inv} (10^4 CNY)	E_m (10^4 CNY)	F_{loss} (10^4 CNY)	Rationale
S0 Baseline	PV: $R_c = R_r$, $T_c = 25$ °C; WT: $v \approx 0.8v_r$; ES: real-time	1175	51.0	2.50	Optimal configuration at typical daily operating point
S1 PV-cloudy	PV: $R_c = 0.6R_r$ (cloudy sky), other parameters are the same as S0	1175	58.5	2.85	PV output decreases → grid-side compensation is required → power outages and grid losses increase
S2 PV-winter	PV: $R_c = 0.75R_r$, $T_c = 5$ °C, other parameters are the same as S0	1175	54.0	2.65	The radiation is weakened but the temperature is reduced to suppress the temperature rise loss, and the impact is moderate
S3 WT-high	WT: $v = 0.9v_r$ (high wind), other parameters are the same as S0	1175	48.0	2.30	Increased wind power → Enhanced local supply → Reduced power outages and grid losses
S4 WT-low	WT: $v = 0.6v_r$ (low wind), other parameters are the same as S0	1175	61.0	3.20	Wind power weakens → tidal backflow increases → network losses rise significantly
S5 ES-ND	ES strategy: night charge (0–6 h)/day discharge (10–16 h), other similarities to S0	1175	52.5	2.55	Fixed-period strategies are slightly weaker than real-time strategies, and peak-valley alignment is insufficient.
S6 ES-RT	ES real-time (voltage/marginal loss trigger), others are the same as S0	1175	49.5	2.35	On-demand scheduling → better voltage and loss characteristics

Table 7 presents the results of the re-optimization sensitivity analysis. Access at the end (Bus 16) is more expensive in terms of power flow. To meet voltage and reliability constraints, optimization tends to slightly increase ES capacity and slightly reduce PV/WT capacity to alleviate local voltage and short-circuit margin pressures. This results in an increase in F_{inv} and a slight increase in F_{loss} and E_m in the same direction. The trunk (Bus 10) maintains minimal losses and good reliability, while the middle section (Bus 6) performs in the middle. This trend is consistent with the power flow patterns of the feeder voltage profile and branch resistance distribution.

Table 7. Re-optimization sensitivity.

ES Bus	ES Size/MW	PV Size/MW	WT Size/MW	F_{inv} (10^4 CNY)	E_m (10^4 CNY)	F_{loss} (10^4 CNY)	Note
10 (main)	0.200	0.570	0.370	1175	51.0	2.50	Baseline compromise solution (trunk injection)
16 (end)	0.240	0.540	0.360	1190	53.0	2.90	The terminal connection requires a larger ES to suppress the voltage drop/loss
6 (mid)	0.215	0.560	0.365	1182	51.8	2.62	The mid-section layout is a compromise between performance and cost

6. Conclusions

This study proposes an improved NSGA-II for distribution network optimization, which enhances convergence speed and solution diversity compared to the standard NSGA-II and PSO. The mechanism aims to enhance the resilience and efficiency of the distribution system. Simulation verification using the IEEE 33-node system demonstrates that the multi-objective optimization framework successfully integrates key factors, including device investment, power supply reliability, and line loss indicators, ensuring an effective balance between these aspects.

The framework incorporates an improved NSGA-II intelligent algorithm, which enables the collaborative regulation of the grid-based distribution network under abnormal operating conditions. This optimization method not only addresses real-time challenges but also enhances operational flexibility. Experimental data confirms that the proposed method strikes a significant balance between operational economy and power supply reliability, reducing system costs while maintaining high levels of service.

Sensitivity results show that although different operating points and access locations can cause fluctuations in economy (F_{inv} only changes during reoptimization), reliability (E_m), and network loss (F_{loss}), the proposed framework maintains a cost–reliability–stability balance across multiple scenarios, demonstrating good adaptability and robustness. The verification results further indicate that the collaborative response scheme significantly improves power supply reliability, emphasizing the advantages of utilizing low-cost DG. It enhances the overall operational efficiency of the distribution network, ensuring a more reliable, cost-effective solution that better meets users' electricity demands. Nevertheless, challenges such as renewable intermittency, demand-side uncertainty, multi-resource coordination, and scalability of optimization algorithms remain key issues for future research. Additionally, this approach can be applied to large-scale grids, improving long-term sustainability. Future work will test the model in real distribution grids. Scenario constraints and N-1 security checks will also be added to improve practical use.

Future research roadmap: the proposed work can be further extended in the following directions:

- (1) Scalability verification: Applying the method to larger-scale distribution networks to assess its scalability and computational performance in more complex scenarios.
- (2) Algorithmic enhancement: Integrating the framework with hybrid optimization algorithms or advanced approaches (e.g., deep reinforcement learning) to improve convergence speed and global search capability.
- (3) Demand–response integration: Incorporating demand–response mechanisms and real-time scheduling strategies to enhance adaptability and robustness under fluctuating loads and renewable-generation variations.

- (4) Engineering validation: Implementing the proposed approach in real-world distribution-network projects for field demonstration and operational verification. In practice, the method can use standard grid operation data such as load profiles and renewable output, and its computational complexity remains manageable. The algorithm can also be extended to larger grids through parallel evaluation, which supports scalability.

Author Contributions: Conceptualization, R.H. and J.H.; methodology, R.H. and J.H.; software, J.H. and F.C.; validation, R.H., J.H. and H.Z.; formal analysis, R.H. All authors have read and agreed to the published version of the manuscript.

Funding: This research is funded by the Research and Demonstration of Grid Collaborative and Mutual Aid Operation Technology for Active Distribution Network with Main Distribution Coordination (030900KC23070002).

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author(s).

Conflicts of Interest: Authors Runquan He, Jiayin Hao, and Heng Zhou were employed by the company Guangdong Power Grid Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Zhao, P.; Wang, J.; Sun, L.; Li, Y.; Xia, H.; He, W. Optimal electrode configuration and system design of compactly-assembled industrial alkaline water electrolyzer. *Energy Convers. Manag.* **2023**, *299*, 117875. [\[CrossRef\]](#)
2. Cheng, Z.; Yan, Y.; Zhou, E.; Gao, B.; Qin, G.; Zhang, K.; Chen, T.; Ma, G.; Huang, X. Synergistic Optimization of Pore and Conductive Network of Short-Cut Graphene Porous Fibers for Lightweight Broadband Electromagnetic Wave Absorption. *Small* **2025**, *21*, e05866. [\[CrossRef\]](#) [\[PubMed\]](#)
3. Zhu, S.; Tian, X.; Chen, H.; Zhu, H.; Qiao, R. Edge collaborative caching solution based on improved NSGA II algorithm in Internet of Vehicles. *Comput. Netw.* **2024**, *244*, 110307. [\[CrossRef\]](#)
4. Liu, Z.; Su, J.; Du, S.; Wang, J.; Zhang, W. A Novel Cooperative Dynamic Optimization Method of Network Reconfiguration and DG Scheduling in Active Distribution Network. *J. Electr. Eng. Technol.* **2025**, 1–14. [\[CrossRef\]](#)
5. Jiang, T.; Guan, Z.; Li, H.; Zhang, M.; Mu, S.; Wu, C.; Jin, M. Collaborative optimization method of cleaning operational performance and multiparameter online control system for combine harvesters. *Comput. Electron. Agric.* **2025**, *235*, 110389. [\[CrossRef\]](#)
6. Tian, Y.; Xu, Y.; Sun, H.; Tai, N. Frequency-constrained microgrid-distribution network coordinated load restoration: A distributed carbon-aware optimization approach. *Appl. Energy* **2025**, *395*, 126221. [\[CrossRef\]](#)
7. Chen, Y.; Bao, Z.; Tan, Y.; Wang, J.; Liu, Y.; Sang, H.; Yuan, X. Hierarchical Charging Scheduling Strategy for Electric Vehicles Based on NSGA-II. *Energies* **2025**, *18*, 3269. [\[CrossRef\]](#)
8. Sun, S.; Yu, P.; Xing, J.; Wang, Y.; Yang, S. Multi-objective collaborative optimization of active distribution network operation based on improved particle swarm optimization algorithm. *Sci. Rep.* **2025**, *15*, 8999. [\[CrossRef\]](#)
9. Li, Y.; Li, K.; Fan, R.; Chen, J.; Zhao, Y. Multi-objective planning of distribution network based on distributionally robust model predictive control. *Front. Energy Res.* **2024**, *12*, 1478040. [\[CrossRef\]](#)
10. Liu, Z.; Chen, Y.; Zhuo, R.; Jia, H. Energy storage capacity optimization for autonomy microgrid considering CHP and EV scheduling. *Appl. Energy* **2018**, *210*, 1113–1125. [\[CrossRef\]](#)
11. Tong, X.; Ma, Q.; Tang, K.; Liu, H.; Li, C. Influence of electric vehicle access mode on the static voltage stability margin and accommodated capacity of the distribution network. *J. Eng.* **2019**, *2019*, 2658–2662. [\[CrossRef\]](#)
12. Kumar, M.; Kumar, A.; Soomro, A.M.; Baloch, M.; Chaudhary, S.T.; Shaikh, M.A. A Comprehensive Review of Optimizing Multi-Energy Multi-Objective Distribution Systems with Electric Vehicle Charging Stations. *World Electr. Veh. J.* **2024**, *15*, 523. [\[CrossRef\]](#)
13. Su, R.; He, G.; Su, S.; Duan, Y.; Cheng, J.; Chen, H.; Wang, K.; Zhang, C. Optimal placement and capacity sizing of energy storage systems via NSGA-II in active distribution network. *Front. Energy Res.* **2023**, *10*, 1073194. [\[CrossRef\]](#)
14. Zhang, Y.; Cui, Q.; Shi, L.; Weng, Y.; Li, J. A Safe energy policy optimization method for multienergy microgrid control. *IEEE Trans. Power Syst.* **2025**, *40*, 1686–1698. [\[CrossRef\]](#)

15. Li, J.; Shi, L.; Fu, H. Multi-objective short-term optimal dispatching of cascade hydro–wind–solar–thermal hybrid generation system with pumped storage hydropower. *Energies* **2025**, *17*, 98. [\[CrossRef\]](#)
16. Jensen, M. Reducing the run-time complexity of multiobjective EAs: The NSGA-II and other algorithms. *IEEE Trans. Evol. Comput.* **2003**, *7*, 503–515. [\[CrossRef\]](#)
17. Zhang, J.; Wang, T.; Liao, Z.; Tang, Z.; Pei, Y.; Cui, Q.; Shu, J.; Zheng, W. Optimal operation strategy for distribution network with high-penetration distributed PV based on soft open point and multi-device collaboration. *Energy* **2025**, *325*, 136191. [\[CrossRef\]](#)
18. Zhang, Y.; Li, D.; Jie, G.; Qirui, R.; Yu, H.; Yuxiao, Z.; Zhang, H. A Cooperative Operation Optimization Method for Medium-and Low-Voltage Distribution Networks Considering Flexible Interconnected Distribution Substation Areas. *Processes* **2025**, *13*, 1123. [\[CrossRef\]](#)
19. Boubaker, S.; Kraiem, H.; Ghazouani, N.; Kamel, S.; Mellit, A.; Alsubaei, F.S.; Bourennani, F.; Meskine, W.; Alqubaysi, T. Multi-objective optimization framework for electric vehicle charging and discharging scheduling in distribution networks using the red deer algorithm. *Sci. Rep.* **2025**, *15*, 13343. [\[CrossRef\]](#)
20. Baptista, J.E.R.; Rodrigues, A.B.; Silva, M.D.G. Probabilistic Analysis of PV generation impacts on voltage sags in LV distribution networks considering failure rates dependent on feeder loading. *IEEE Trans. Sustain. Energy* **2018**, *10*, 1342–1350. [\[CrossRef\]](#)
21. Li, S.; Shao, J.; Zhang, S.; Chen, H. Distribution network reconfiguration incorporating thermostatic loads after multiple line contingency. *Energy Rep.* **2025**, *13*, 813–823. [\[CrossRef\]](#)
22. Heris, S.M.K.; Khaloozadeh, H. Open- and closed-loop multiobjective optimal strategies for HIV therapy using NSGA-II. *IEEE Trans. Biomed. Eng.* **2011**, *58*, 1678–1685. [\[CrossRef\]](#) [\[PubMed\]](#)
23. Nie, S.; Cai, J.; Ji, H.; Zhang, J.; Hong, R. Multi-objective optimization integrating weighted average surrogate model and NSGA-II intelligent algorithm applied to a self-excited oscillation mixer used in mixed flow descaling. *Proc. Inst. Mech. Eng. Part C J. Mech. Eng. Sci.* **2024**, *238*, 7444–7462. [\[CrossRef\]](#)
24. Yang, X.; Long, J.; Liu, P.; Zhang, X.; Liu, X. Optimal scheduling of microgrid with distributed power based on water cycle algorithm. *Energies* **2018**, *11*, 2381. [\[CrossRef\]](#)
25. Gbadega, P.A.; Sun, Y.; Balogun, O.A. A Unified Optimization Framework for Cost-Effective and Voltage-Stable Operation of Renewable Energy-Based Microgrids Using Mixed-Integer Nonlinear Programming. *e-Prime* **2025**, *12*, 101010. [\[CrossRef\]](#)
26. Xie, H.; Lin, G.; Wu, H.; Tong, Z.; He, Y.; Chen, L.; Liu, C.; Jiang, T. Sensing Performance of Cr_n (*n* = 1–3) Clusters Doped WSe₂ and WTe₂ for Gases in Power Transformer Winding Deformation Fault by DFT Calculation. *Langmuir* **2025**, *41*, 8781–8803. [\[CrossRef\]](#)
27. Pang, H.; Wang, J.; Hu, X. Understanding the potential influence of wechat engagement on bonding capital, bridging capital, and electronic word-of-mouth intention. *Sustainability* **2021**, *13*, 8489. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.